

Final Project - Step 2 (15 Points)

PSTAT100: Data Science Concepts and Analysis

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1 Step 2: Research Questions, Hypotheses, and Exploratory Data Analysis (EDA)

1.1 Research Questions

Q1: Does family income play a role in dropout rates after 12th grade?

Q2: Does the simulated test score accurately predict the JEE (Joint Entrance Examination) score that the class 12 students receive?

Q3: Are there significant differences in study habits (study hours) between students who drop out and those who continue.

1.2 Hypotheses

(1) Family income will play a significant role in impacting a student's dropout rate.

(2) Based on the simulated JEE test score, the true score that the students receive on the exam should reflect the mock test results.

(3) Students who continue to study significantly more hours per day than students who drop out.

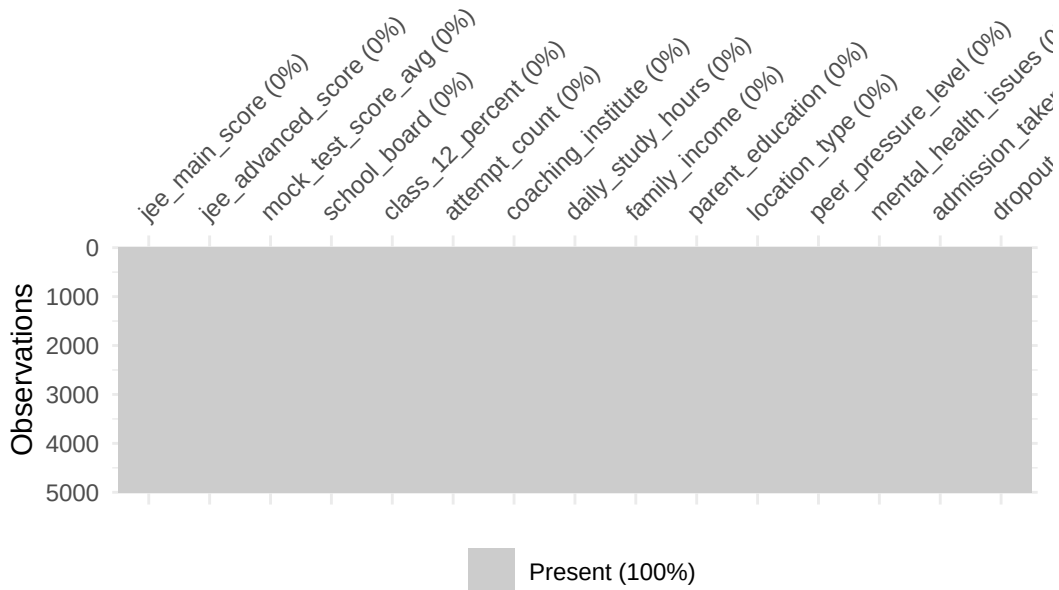
1.3 Exploratory Data Analysis (EDA)

1.4 Data Cleaning

```
1 colSums(is.na(data))
```

jee_main_score	jee_advanced_score	mock_test_score_avg
0	0	0
school_board	class_12_percent	attempt_count
0	0	0
coaching_institute	daily_study_hours	family_income
0	0	0
parent_education	location_type	peer_pressure_level
0	0	0
mental_health_issues	admission_taken	dropout
0	0	0

```
vis_miss(data)
```



Since there is no missing data, we do not need to further drop or adjust any row. Thus, we will conduct tests on this dataset. However, we still further clean data to factor to smooth the plots.

1.5 Descriptive Statistics

```
summary(data)
```

jee_main_score	jee_advanced_score	mock_test_score_avg	school_board
Min. : 26.62	Min. : 3.24	Min. : 27.49	Length:5000
1st Qu.: 62.79	1st Qu.: 55.02	1st Qu.: 60.07	Class :character
Median : 72.19	Median : 65.72	Median : 68.12	Mode :character
Mean : 71.96	Mean : 65.74	Mean : 68.11	
3rd Qu.: 81.32	3rd Qu.: 76.83	3rd Qu.: 76.11	
Max. :100.00	Max. :100.00	Max. :100.00	
class_12_percent	attempt_count	coaching_institute	daily_study_hours
Min. : 50.00	Min. :1.000	Length:5000	Min. : 0.000
1st Qu.: 68.07	1st Qu.:1.000	Class :character	1st Qu.: 3.100
Median : 74.94	Median :2.000	Mode :character	Median : 4.500
Mean : 74.96	Mean :1.505		Mean : 4.484
3rd Qu.: 81.84	3rd Qu.:2.000		3rd Qu.: 5.900
Max. :100.00	Max. :2.000		Max. :10.000

family_income	parent_education	location_type	peer_pressure_level
Length:5000	Length:5000	Length:5000	Length:5000
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

mental_health_issues	admission_taken	dropout
Length:5000	Length:5000	Min. :0.0000
Class :character	Class :character	1st Qu.:0.0000
Mode :character	Mode :character	Median :0.0000
		Mean :0.2072
		3rd Qu.:0.0000
		Max. :1.0000

```
1 data_clean <- data %>%
2   mutate(across(c(school_board, coaching_institute, family_income,
3                   parent_education, location_type, peer_pressure_level,
4                   mental_health_issues, admission_taken), as.factor),
5           dropout = as.factor(dropout))
```

The data_clean dataset has 5,000 rows and 15 columns. It includes both numerical and categorical (factor) variables. The JEE Main score, JEE Advanced score, mock test score average, Class 12 percentage, attempt count, and daily study hours are numeric variables, meaning they contain numbers. The other columns are categorical (factor) variables. These include school board, coaching institute, family income, parent education, location type, peer pressure level, mental health issues, admission taken, and dropout. Each of these has a set number of categories or levels. For example, “school board” has 3 levels (CBSE, ICSE, and others), and “mental health issues” has 2 levels (“Yes” or “No”). The “dropout” column shows whether a student dropped out or not and is also treated as a factor with two levels (“0” = No, “1” = Yes).

```
1 data %>%
2   group_by(parent_education) %>%
3   summarise(
4     count = n(),
5     proportion = n() / nrow(data))
```

```
# A tibble: 4 x 3
  parent_education count proportion
  <chr>          <int>     <dbl>
1 12th           1243     0.249
2 Graduate       1224     0.245
3 PG             1245     0.249
4 Upto 10th      1288     0.258
```

```
1 data %>%
2   group_by(family_income) %>%
3   summarise(
4     count = n(),
5     proportion = n() / nrow(data))
```

```
# A tibble: 3 x 3
  family_income count proportion
  <chr>          <int>     <dbl>
```

1 High	1663	0.333
2 Low	1679	0.336
3 Mid	1658	0.332

For parent education, the proportion of parents who only have a high school education is 24.86%. Parents who attended graduate school are 24.48%. Those who attended post-graduate school are 24.9% and those who only attended up to 10th grade are 25.76% Most of the parents have only attended up to 10th grade.

In terms of family income, the proportion of those who have a high income is 33.26%, , mid income is 33.16%, and low income is 33.58%. Most of the students' family income is low, but not by a large margin.

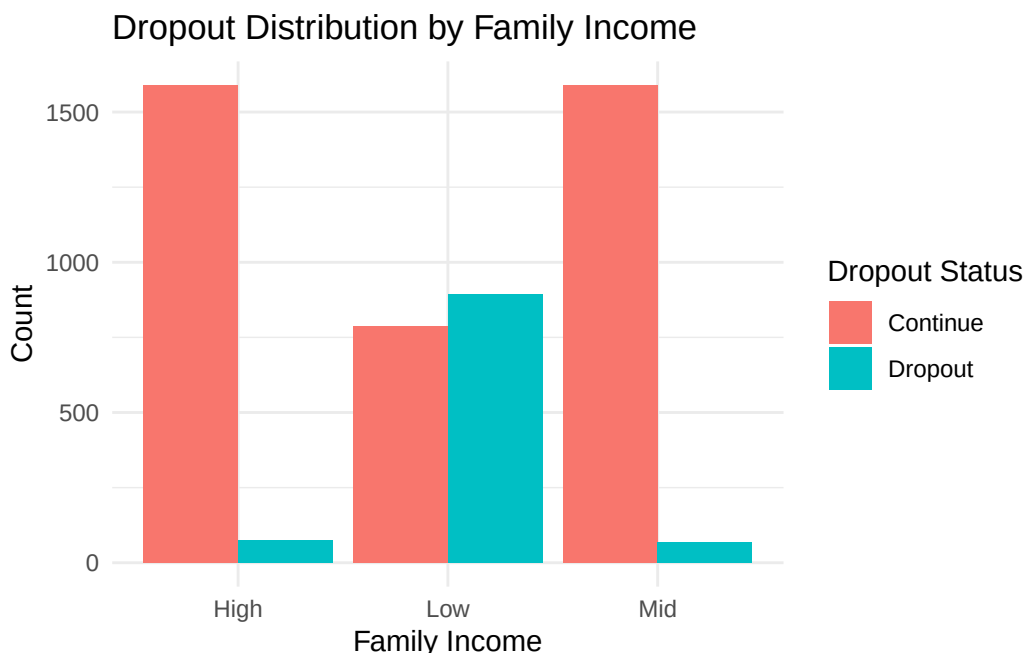
1.6 Data Visualization

1.6.1 Hypothesis 1:

```

1 # Make sure dropout is a factor for colors
2 data$dropout <- factor(data$dropout, labels = c("Continue", "Dropout"))
3
4
5 # 1. Histogram of family income by dropout status
6 p1 <- ggplot(data, aes(x = family_income, fill = dropout)) +
7   geom_bar(position = "dodge") +
8   labs(x = "Family Income", y = "Count", fill = "Dropout Status") +
9   ggtitle("Dropout Distribution by Family Income") +
10  theme_minimal()
11 p1

```



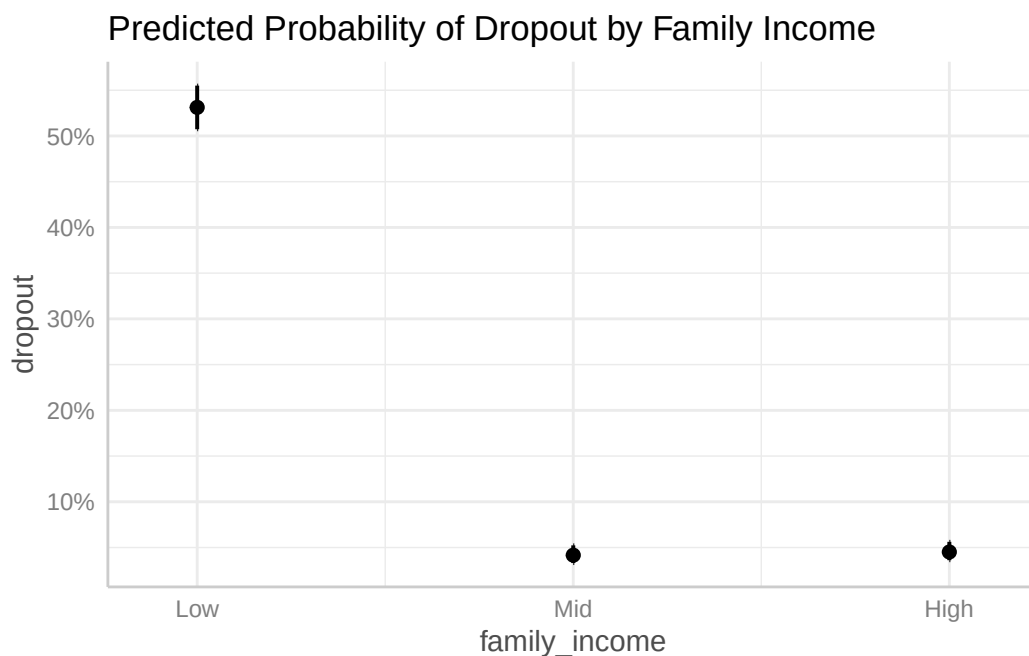
Looking at the box plot visualizing the dropout rates by family income, it is clear that family income has a strong influence. We used box plots to compare how dropout proportions differ from to group to group.

For the family income box plot, low-income families have a much higher dropout rate while dropout rates for mid and high income groups are very low. It seems that more than 50% of the students whose families are low income have dropped out while those who come from mid to low income families have close to 0%.

```

1 model <- glm(dropout ~ family_income, data = data, family = "binomial")
2
3 plot(ggpredict(model, terms = "family_income")) +
4   ggtitle("Predicted Probability of Dropout by Family Income")

```



We used logistic regression to estimate the probability that a student will drop out after 12th grade based on family income level. For those with low family income, the probability that they will drop out is very high. It seems that more than half of the students from low-income families are predicted to drop out. Students with a mid to high dropout rate have a substantially lower risk.

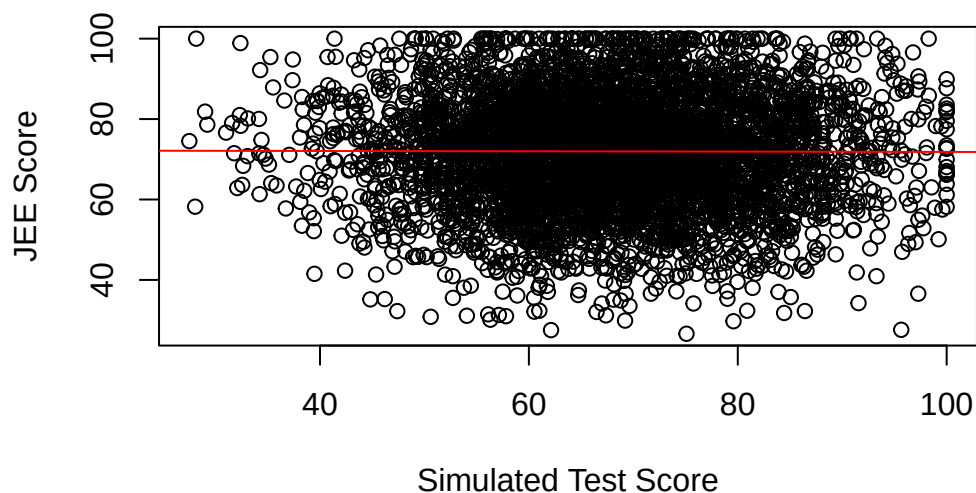
1.6.2 Hypothesis 2:

```

1 # Visualize relationship between simulated and true score
2 plot(data$mock_test_score_avg, data$jee_main_score,
3       main = "Simulated Score vs JEE Score",
4       xlab = "Simulated Test Score",
5       ylab = "JEE Score")
6 abline(lm(jee_main_score ~ mock_test_score_avg, data = data), col = "red")

```

Simulated Score vs JEE Score



```

1 # Correlation check
2 cor(data$mock_test_score_avg, data$jee_main_score)

```

```
[1] -0.003406336
```

A correlation of -0.0034 between the simulated test score and JEE score indicates there is almost no linear relationship between the two variables.

```

1 # Linear regression
2 model <- lm(jee_main_score ~ mock_test_score_avg, data = data)
3 summary(model)

```

Call:

```
lm(formula = jee_main_score ~ mock_test_score_avg, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-45.314	-9.177	0.226	9.373	28.156

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	72.22685	1.11981	64.499	<2e-16 ***
mock_test_score_avg	-0.00390	0.01619	-0.241	0.81

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.67 on 4998 degrees of freedom

Multiple R-squared: 1.16e-05, Adjusted R-squared: -0.0001885

F-statistic: 0.05799 on 1 and 4998 DF, p-value: 0.8097

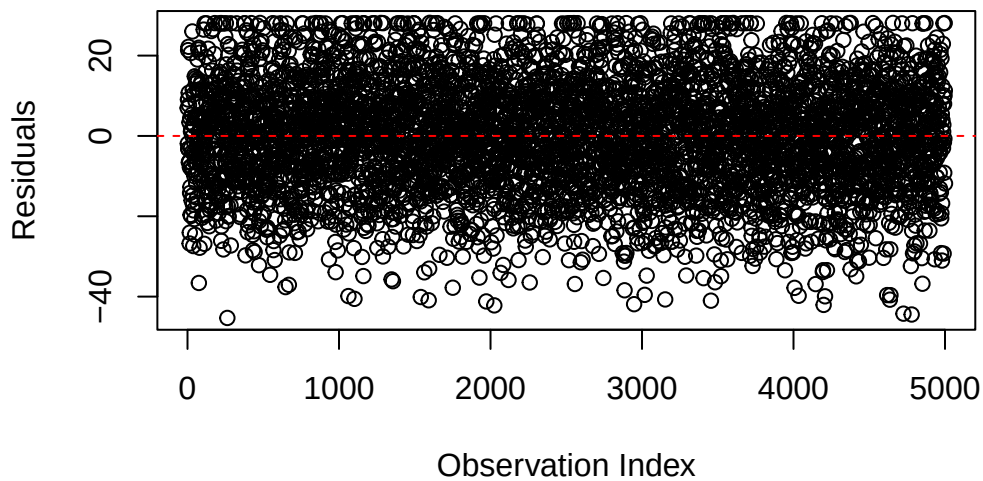
Based on the regression summary, it appears that the simulated scores do not significantly predict JEE Main scores. The model appears to be statistically insignificant as the p-value is 0.8097 and the R-squared is incredibly low, so it does not explain much of the variation in the data. Additionally, the low F-statistic demonstrates that there is little evidence that the predictor adds value to the model.

```

1 plot(model$residuals,
2       main = "Residual Plot",
3       xlab = "Observation Index",
4       ylab = "Residuals")
5 abline(h = 0, col = "red", lty = 2)

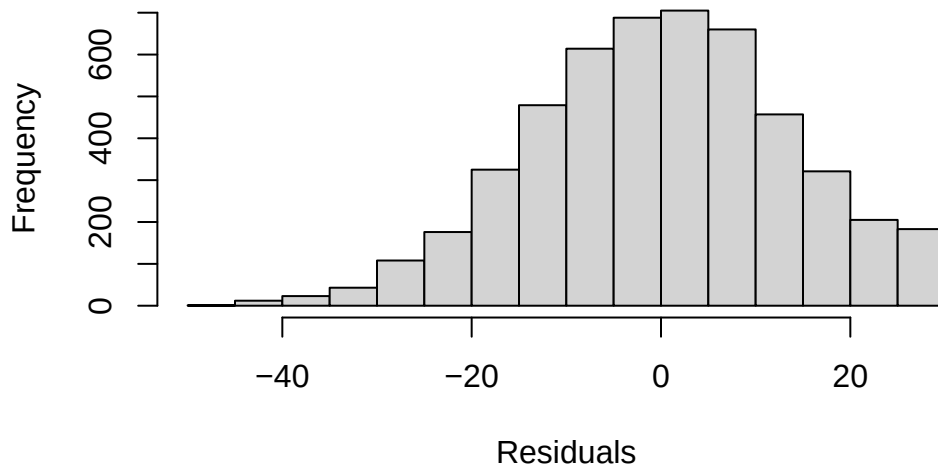
```

Residual Plot



```
1 hist(model$residuals,  
2     main = "Histogram of Residuals",  
3     xlab = "Residuals",  
4     ylab = "Frequency",  
5     border = "black")
```

Histogram of Residuals



1.6.3 Hypothesis 3:

H_0 : There is no significant difference in daily study hours between students who continue their education and those who drop out.

H_1 : Students who continue their education study significantly more hours per day than students who drop out.

```
1 # using Welch's t-test, which does not use the standard n1 + n2 - 2 formula for degrees of freedom.  
2 # Instead, it uses the Welch-Satterthwaite approximation, which adjusts the df based on:  
3 # 1) Unequal group sizes  
4 # 2) Unequal variances
```

Given the groups (3964 vs. 1036), and likely different variances in hours studied, Welch's formula calculates an adjusted, lower df — in your case, ~1454 — to ensure the test remains conservative and accurate

```

1 continue <- data_clean$daily_study_hours[data_clean$dropout == 0]
2 dropout <- data_clean$daily_study_hours[data_clean$dropout == 1]
3 t_test_result <- t.test(continue, dropout, alternative = "greater", var.equal = FALSE)
4 print(t_test_result)

```

Welch Two Sample t-test

```

data: continue and dropout
t = 8.2004, df = 1453.8, p-value = 2.604e-16
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 0.4935631      Inf
sample estimates:
mean of x mean of y
 4.611806  3.994305

```

Since the p-value (=2.604e-16) is far less than 0.05, the null hypothesis is rejected.

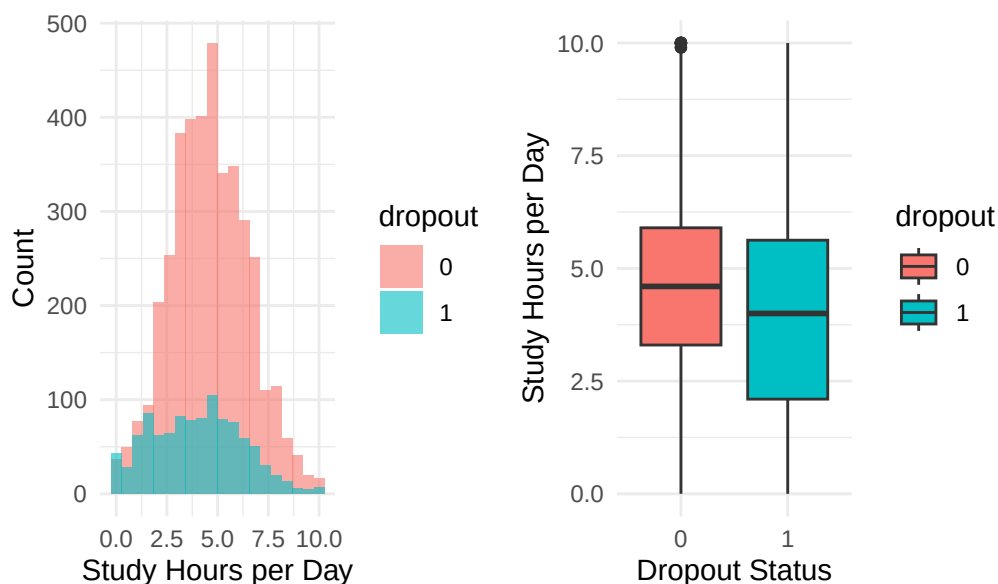
Conclusion: There is strong statistical evidence that students who continue their studies after Class 12 study significantly more hours per day than those who drop out.

```

1 plot1 <- ggplot(data_clean, aes(x = daily_study_hours, fill = dropout)) +
2   geom_histogram(alpha = 0.6, position = "identity", bins = 20) +
3   labs(x = "Study Hours per Day",
4        y = "Count") +
5   theme_minimal()
6
7 plot2 <- ggplot(data_clean, aes(x = dropout, y = daily_study_hours, fill = dropout)) +
8   geom_boxplot() +
9   labs(x = "Dropout Status",
10        y = "Study Hours per Day") +
11   theme_minimal()
12
13 combined_plots <- (plot1 + plot2) +
14   plot_annotation(title = "Study Hours Distribution and Boxplot by Dropout Status")
15 combined_plots

```

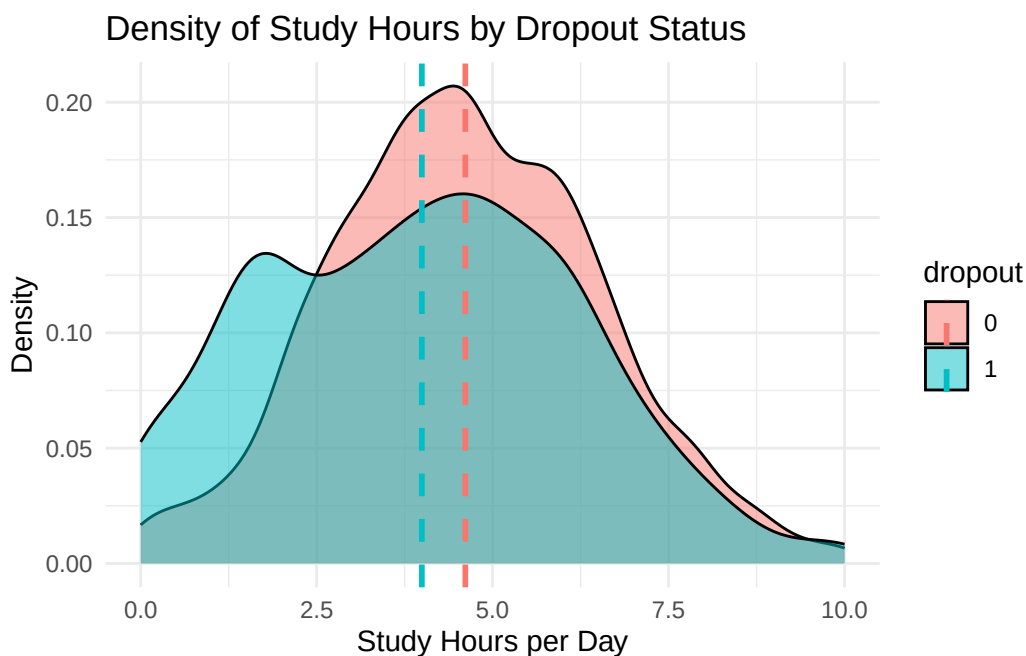
Study Hours Distribution and Boxplot by Dropout Status



The histogram suggests that the daily study hour in both continued and dropout students are approximately normally distributed. The mean study hour for continued student is slightly higher than the dropout student, but the difference is small. Moreover, the distribution for continued student are more concentrated toward the center, whereas the distribution for dropout student is more spread out. This observation is confirmed by the boxplot, which also suggests that the median study hour for continued student is slightly higher than dropout student, and the spread of the distribution is smaller compares to the dropout student.

However, as shown in the histogram, there is an uneven number of data points between dropout and continued students. There are far more continued student than dropout students, which makes it very hard to tell the distribution of the dropout student.

```
1 ggplot(data_clean, aes(x = daily_study_hours, fill = dropout)) +
2   geom_density(alpha = 0.5) +
3   labs(title = "Density of Study Hours by Dropout Status",
4         x = "Study Hours per Day",
5         y = "Density") +
6   theme_minimal() +
7   geom_vline(data = data_clean %>%
8             group_by(dropout) %>%
9             summarise(mean_hours = mean(daily_study_hours)),
10             aes(xintercept = mean_hours, color = dropout),
11             linetype = "dashed", size = 1)
```



Hence, we plot the histogram again in relative density scale, where the uneven sample size is no longer a problem. We see a similar structure with the boxplot. The study hour distribution for dropout student is more spread out and has a mean slightly toward the left. We also notice a new structure that is not shown in the boxplot: a secondary peak in the dropout distribution. This suggests that the study hour for dropout students are more likely to follow a bi-modal distribution with one of the peaks at around 4.6 hours (same with continued students) and the other peak at 1.75 hour. This reflectst that there might be two types of dropout students: The first type study as much as continued students, while the second type study significantly less than the continued students.